
Chapter - 3 COST

3.1 Cost classification

3.1.1 Total, average, fixed, variable, and marginal costs

3.1.2 Direct, indirect, and standard costs

3.1.3 Cash versus book cost, manufacturing and non-manufacturing cost

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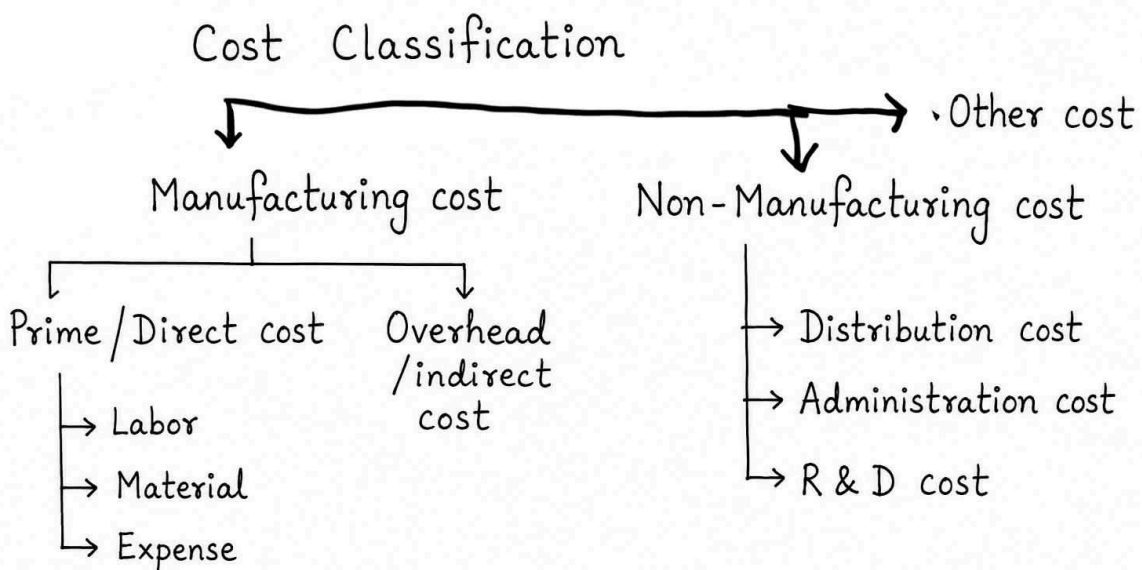
3.2 Cost estimation techniques Cost control (NUM)

[ALL Portion Are Theoretical Except 3.2]

3.1 Cost classification

Cost is the monetary value of resources sacrificed to obtain a product, service, or objective.

Cost classification is the process of grouping costs according to their nature, behavior, function, or relevance for decision-making.



3.1.1 Total, average, fixed, variable, and marginal costs

Total Cost (TC) - Total cost is the total expenditure incurred in producing a certain quantity of goods or services.

$$TC = \text{Total Fixed Cost} + \text{Total Variable Cost}$$

Average Cost (AC) - Average cost is the cost per unit of output.

$$AC = TC / \text{Quantity-Produced}$$

Fixed Cost (FC) - Cost that doesn't vary with production volume

Variable Cost (VC) - Cost that varies with production volume

Marginal Cost (MC) / Incremental Cost (IC) - additional cost of producing one extra unit.

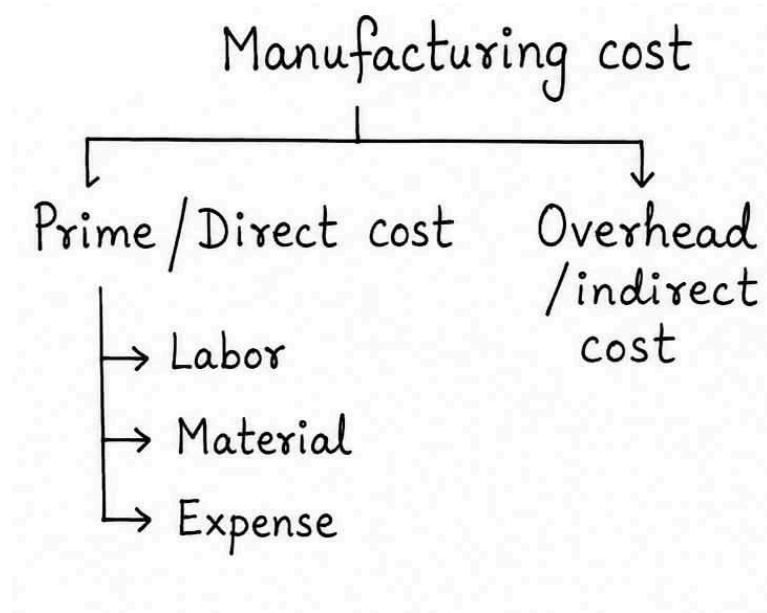
$$MC = \Delta TC / \Delta Q$$

3.1.2 Manufacturing Costs

Direct Cost (DC) / Prime Cost (PC) - Cost that can be easily / economically traced

Indirect Cost (IC) / Overhead Cost (O&H) - Cost that can't be traced

Standard Cost / Estimation Cost - Predetermined estimated cost established under normal operating conditions. It serves as a benchmark for cost control.



3.1.3 Cash versus book cost

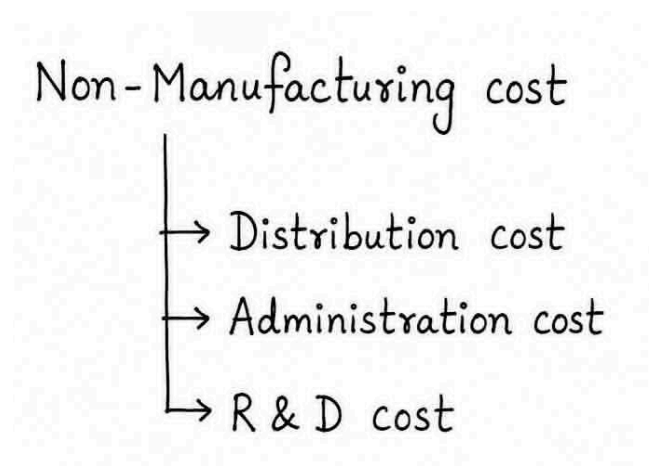
Cash cost and book cost are two different concepts used in accounting and financial analysis.

Aspect	Cash Cost	Book Cost
Definition	Costs that involve an actual cash outflow during the period.	Total cost recorded in the accounting books, including both cash and non-cash expenses.
Cash Payment	Yes	May or may not involve cash payment.
Includes	Raw materials, wages, salaries, rent, utilities, interest paid, etc.	All cash costs plus non-cash expenses like depreciation, amortization, and provisions.
Excludes	Non-cash expenses such as depreciation.	Excludes only items not recognized as expenses.
Purpose	Used to assess liquidity and cash flow requirements.	Used to determine accounting profit and financial reporting.

Formula

- $\text{Cash Cost} = \text{Total Cost} - \text{Non-cash Expenses}$
- $\text{Book Cost} = \text{Cash Costs} + \text{Non-cash Expenses}$

3.1.4 Non-manufacturing cost



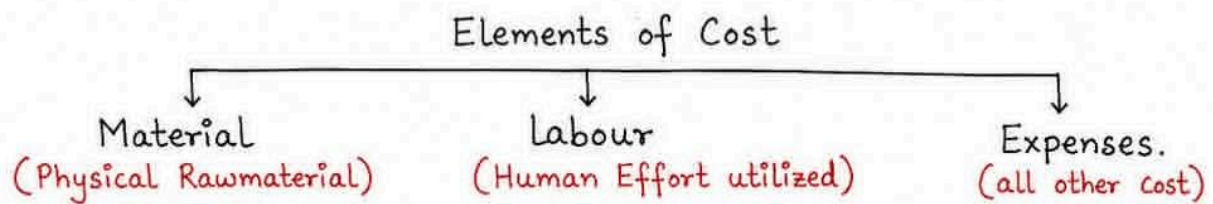
3.1.5 Other Production Cost

Sunk cost - Cost that has already been incurred and cannot be recovered

Opportunity cost - Value of the next best alternative that is sacrificed

Decision Rule: Always ignore sunk costs **when choosing between future alternatives.**

3.1.6 Element of cost



3.1.7 Life-cycle cost (LLC) / Womb-to-Tomb Cost / Cradle-to-Grave Cost / Initial-to-Final Cost / Whole-Life Cost / Total Cost Of Ownership

Life-cycle cost is the total cost incurred throughout the entire life of a product or asset, from planning and design to disposal.

$$\text{LCC} = \text{Initial Cost} + \text{Operating Cost} + \text{Maintenance Cost} + \text{Disposal Cost}$$

Purpose / Objective / Why To Do It ?

Helps select the most economical alternative.

Improves long-term investment decisions.

Reduces total ownership cost.

Supports budgeting and maintenance planning.

Important Exam Definitions

- Total Cost: Sum of fixed and variable costs.
- Average Cost: Cost per unit of output.
- Fixed Cost: Cost that does not change with production.
- Variable Cost: Cost that varies with production.
- Marginal Cost: Additional cost of producing one more unit.
- Direct Cost: Cost directly traceable to a product.
- Indirect Cost: Cost not directly traceable to a product.
- Standard Cost: Predetermined expected cost used as a benchmark.
- Cash Cost: Cost involving actual cash payment.
- Book Cost: Non-cash accounting cost.
- Manufacturing Cost: Cost incurred in producing goods.
- Non-Manufacturing Cost: Selling and administrative costs.
- Sunk Cost: Past irrecoverable cost.
- Opportunity Cost: Value of the next best alternative forgone.
- Life-Cycle Cost: Total cost of an asset from acquisition to disposal.

[2082 Chaitra]

An engineering firm is choosing between two types of road pavement materials:

Material A: High initial cost but low annual maintenance cost.

Material B: Low initial cost but frequent, expensive repairs.

- (i) Which cost classification should the engineer use to compare these two?
- (ii) Explain the importance of Life-Cycle Costing (LCC) analysis in selecting the most economic design alternative.

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(i) The appropriate cost classification is Life-Cycle Costing (LCC) or Whole-Life Cost analysis, which compares alternatives based on their total cost over the entire service life, including initial, maintenance, repair, and disposal costs.

(ii) Life-Cycle Costing is important because it:

1. considers all costs over the pavement's life,
2. compares alternatives on total economic cost rather than only initial cost,
3. helps select the most cost-effective design,
4. reduces future maintenance and repair expenses,
5. improves long-term value and resource allocation, and
6. accounts for the time value of money by discounting future costs to present value.

Therefore, Material A may be the better choice despite its higher initial cost if its total life-cycle cost is lower than that of Material B.

3.2 Cost estimation techniques Cost control (NUM)

Cost estimation is the process of predicting the cost of manufacturing a product or completing a project before actual production begins.

3.2.1 Variance Analysis:

$$\text{Total Variance} = \text{Actual Cost} - \text{Standard Cost}$$

If Actual Cost > Standard Cost → Unfavorable Variance

If Actual Cost < Standard Cost → Favorable Variance

3.2.2 Break Even Analysis

$$\text{BreakEven-Volume (Q)} = (\text{Fixed Cost} + \text{Desired Profit}) / \text{SP} - \text{VC}$$

3.2.3 High-Low Activity Analysis

Variable Cost per Unit

$$VC = \frac{\text{Highest Cost} - \text{Lowest Cost}}{\text{Highest Activity} - \text{Lowest Activity}}$$

Fixed Cost

$$FC = \text{Total Cost} - (VC \times \text{Activity})$$

3.3 Cost Control

Cost control is the process of ensuring that actual costs remain within predetermined standards or budgets. Steps in Cost Control are as follows :

1. Set standards
2. Measure actual cost
3. Compare with standards
4. Find variance
5. Correct deviations

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